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Steering column mount bracket release feature

Abstract

A steering column assembly has a mount bracket configured for attachment to a fixed vehicle support member in a releasably fixed position and for release from the fixed vehicle support member to a released position during a steering column energy absorbing event. A steering jacket is operably supported by the mount bracket. The steering jacket has at least one interface feature arranged to confront the mount bracket during the steering column energy absorbing event to inhibit rotational movement of the mount bracket relative to the steering jacket when the mount bracket moves from the fixed position to the released position.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10518799	12/2018	Nagatani	N/A	B62D 1/185
2007/0228717	12/2006	Tanai	280/777	F16F 7/128
2008/0238070	12/2007	Bodtker	74/492	B62D 1/195

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/545,512, filed Oct. 24, 2023, which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) This disclosure generally relates to steering column assemblies and, more particularly, to steering column assemblies with releasable mount brackets.

BACKGROUND

(2) Automotive steering column assemblies are known to include a mount bracket releasably attached to a vehicle frame member for intentional release during impact of a driver with a steering wheel. Smooth release of the mount bracket from the frame member is desired, without binding between the mount bracket and the vehicle frame member. Unfortunately, the mount bracket can become tilted relative to the vehicle frame member during release, thereby resulting in unwanted resistance and binding between the mount bracket and the vehicle frame member.

SUMMARY

(3) In accordance with one aspect of the disclosure, a steering column assembly includes a mount bracket configured for attachment to a fixed vehicle support member in a releasably fixed position and for release from the fixed vehicle support member to a released position during a steering column energy absorbing event. A steering jacket is operably supported by the mount bracket. The steering jacket has at least one interface feature arranged to confront the mount bracket, during the steering column energy absorbing event, to inhibit rotational movement of the mount bracket relative to the steering jacket when the mount bracket moves from the fixed position to the released position.

(4) In accordance with one aspect of the disclosure, a method of causing a steering jacket to inhibit rotation of a mount plate as the mount plate moves from a releasably fixed position with a fixed vehicle support member to a released position from the fixed vehicle support member during a steering column energy absorbing event, includes: providing the steering jacket having at least one interface feature extending toward the mount plate, with the at least one interface feature arranged to confront the mount plate during the steering column energy absorbing event to inhibit rotational movement of the mount plate relative to the steering jacket when the mount plate moves from the releasably fixed position to the released position.

(5) These and other advantages and features will become more apparent from the following description taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter that is regarded as the invention is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other features, and advantages of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is a perspective view of motor vehicle including a steering column assembly constructed in accordance with an aspect of the disclosure;

(3) FIG. 2 is a fragmentary side view of the steering column assembly of FIG. 1;

(4) FIG. 2A is a perspective view of the steering column assembly of FIG. 2;

(5) FIG. 2B is a fragmentary view looking generally along a longitudinal axis of the steering column assembly toward a mount bracket of the steering column assembly shown spaced by a gap from an interface feature of a steering column jacket of the steering column assembly prior to an energy absorbing event;

(6) FIG. 2C is a view similar to FIG. 2B illustrating the steering column assembly during an energy absorbing event with the interface feature inhibiting rotation of the mount bracket in accordance with the disclosure;

(7) FIG. 3 is a view similar to FIG. 2A with various components removed for clarity purposes of illustrating the interface feature on the steering column jacket constructed in accordance with one aspect of the disclosure; and

(8) FIG. 4 is a view similar to FIG. 3 illustrating an interface feature on a steering column jacket constructed in accordance with another aspect of the disclosure.

DETAILED DESCRIPTION

(9) Referring now to the Figures, the present disclosure will be described with reference to specific steering shaft assembly and clamp yoke embodiments. It is to be understood that the disclosed embodiments are merely illustrative of the present disclosure and may be embodied in various and alternative forms. Various elements of the disclosed embodiments may be combined or omitted to form further embodiments of the present disclosure. The Figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

(10) FIG. 1 illustrates a motor vehicle **10** having a steering column assembly **12** operably coupled to a steering gear assembly **14** by a steering shaft assembly, such as an intermediate shaft assembly, and referred to hereafter as intermediate shaft **16**, constructed in accordance with an aspect of the disclosure. The steering column assembly **12** has a steering jacket assembly including one or more coaxially aligned steering jackets, with an upper steering jacket **18** shown supported by a mount

bracket **20**. The mount bracket **20** is configured for attachment to a fixed vehicle support member **22**, such as vehicle frame member, by way of example and without limitation, in a releasably fixed position, such as via energy absorption straps **23**. The mount bracket **20** is configured for release from the fixed vehicle support member **22**, whereupon mount bracket **20** translates in sliding relation along energy absorption straps **23** with the upper steering jacket **18**, and in accordance with the disclosure, the mount bracket **20** translates without rotation relative to the upper steering jacket **18**, or at least without substantial rotation (meaning that the mount bracket, if it rotates at all, only rotates along the direction of arrow A between about 0.1-5 degrees, and more preferably less than 3 degrees), to a released position during a steering column energy absorbing event. The minimal relative rotation between the upper steering jacket **18** and mount bracket **20** facilitates smooth sliding translation of the mount bracket **20** and upper steering jacket **18** relative to the energy absorption straps **23**, without binding of the mount bracket **20**, thereby absorbing energy and minimizing potential harm to the driver. The energy absorbing event can occur during a crash condition, whereupon a driver of the motor vehicle **10** may impact a steering wheel **24** of the steering column assembly **12**, thereby causing an impact force F, acting generally along a longitudinal axis **26** of the steering column assembly **12**, to move the upper steering jacket **18** axially, generally along the longitudinal axis **26**, toward a front end **28** of the motor vehicle **10**.

(11) The mount bracket **20** of the steering column assembly **10** is a rake type mount bracket, wherein actuation of a lever **30** supported by a bolt **32** releases steering column assembly **12** for tilting movement, as desired by the driver. The driver, upon orienting the steering column assembly **12** in the desired inclined orientation, locks the steering column assembly **12** in place via lever **30**. Mount bracket **20** includes a pair of release capsule slots, referred to hereafter as release slots **31**.

(12) During a steering column energy absorbing event, such as an impact force against the front end **28** of motor vehicle **10**, by way of example and without limitation, force F is transmitted from a compression bracket **34**, which is fixed to upper steering jacket **18**, through the bolt **32** to the mount bracket **20**. Release slots **31** of mount bracket **20** are then caused to separate from vehicle support member **22**, whereupon mount bracket **20** slides smoothly, without tipping, also referred to as rotation, along energy absorption straps **23**.

(13) As mount bracket **20** separates from vehicle support member **22** and begins sliding along energy absorption straps **23**, smooth, generally planar movement, intended to mean that an upper planar surface **20a** of mount bracket **20** translates along and without tipping or without substantial tipping (minimal tipping, such as less than 3 degrees) relative to a plane P is promoted. Plane P is defined being parallel or substantially parallel (intended to mean within 5 degrees or less of true parallel) to the upper planar surface **20a** of mount bracket **20** and parallel or substantially parallel (intended to mean within 5 degrees or less of true parallel) to longitudinal axis A. As such, smooth, non-binding movement of mount plate **20** along energy absorption straps **23** is facilitated via at least one stabilizing feature, also referred to as interface feature **36**, arranged to confront the mount bracket **20**, during the steering column energy absorbing event, to inhibit rotational movement of mount bracket **20** relative to the steering jacket **18** as the mount bracket **20** moves from the fixed position to the released position. As such, reaction loading (binding friction) between mount bracket **20** and release slots **31** is minimized, and thus, the mount bracket **20** is free to release from release slots **31** under the energy absorbing event, such that mount bracket **20** is prevented from deforming and the movement of mount bracket **20** is controlled, as intended, by the energy absorption straps **23**, thereby controlling the energy load profile. As best shown in a non-limiting embodiment of FIG. 3, interface feature **36** is provided as a plurality of interface features **36** fixed to upper steering jacket **18**. The interface features **36** are formed as a monolithic piece of the material of upper steering jacket **18**, and can be formed as plastically deformed material of the upper steering jacket **18**, such as in mechanical material upsetting operation, including stamping, by way of example and without limitation. As such, no additional material is needed in addition to the material of upper steering jacket **18** to form the interface features **36**.

(14) Interface features **36** extend radially outwardly from an outer surface of the upper steering jacket **18**. In the non-limiting embodiment illustrated, the interface features **36** extend radially outwardly from an outer surface **38**, such as a generally straight cylindrical outer surface **38** of upper steering jacket **18**, by way of example and without limitation, toward the mount bracket **20**, with generally intended to mean that the outer surface **38** can be purely straight and purely cylindrical, or slightly less than straight, such as up to about 3 degrees off purely straight, and slightly out of cylindrical contour, such as being non-cylindrical in visual appearance. Accordingly, the interface feature(s) **36** reduce the distance (width) of a gap **G** (FIG. **2B**, looking generally from the steering wheel **24** toward the mount bracket **20** along the longitudinal axis **26**) extending between the upper steering jacket **18** and the mount bracket **20**, and in particular, between a peak of the interface feature(s) **36** and a bottom surface of the mount bracket **20**. Interface features **36** are shown spaced circumferentially from one another, and are equidistantly spaced from an end **40** of the upper steering jacket **18**. Accordingly, as upper steering jacket **18** is moved generally along longitudinal axis **26** in an energy absorbing event (from right to left as viewed in FIG. **2**), interface features **36** confront and engage a rearwardly facing surface **43** of mount bracket **20**, such as extending upwardly from the bottom surface **42** of mount bracket **20**, thereby causing mount bracket **20** and upper steering jacket to move conjointly and simultaneously with one another. The bottom surface **42** is shown as having a radiused scalloped region, referred to hereafter as scallop **44**, contoured to provide clearance of the upper steering jacket **18** with mount bracket **20** in a rake up (uppermost adjustment of steering column assembly **12** during normal use) position of steering column assembly **12**. The interface features **36** are arranged and aligned to confront and abut surface **43** adjacent the region of radiused scallop **44** during the energy absorbing event (FIG. **2C**), thereby causing mount bracket **20** to translate along plane **P** with upper steering jacket **18**. As such, the extent to which the interface features **36** need to extend outwardly from the outer surface **38** of upper steering jacket **18** to engage surface **43** is minimized, such as between about 1-5 mm, and more preferably between about 1-3 mm.

(15) With interface features engaging surface **43** of mount bracket **20** during an energy absorbing event, the force transmitted to mount bracket **20** via bolt **32** is counteracted by a Force **F1** imparted by interface features **42** against the surface **43**, and thus, mount bracket **20** is inhibited from rotational movement downwardly relative to upper steering jacket **18** and vehicle support member **22**. It is contemplated that a very small degree of rotational movement of mount bracket **20** may occur, such as between about 1-5 degrees at most, and preferably less than 3 degrees. Accordingly, movement of mount bracket **20** during the energy absorbing event may be in pure translation with no rotation, or in substantial translation with upper steering jacket **18** without a high degree of rotation, as discussed above.

(16) As shown in another non-limiting embodiment of FIG. **4**, a steering column assembly **112** is shown having an interface feature **136** constructed in accordance with another aspect of the disclosure. The interface feature **136** functions as discussed above for interface features **36**, but rather than being formed as a plurality of interface features spaced from one another, interface feature **136** is formed as a single interface feature. Interface feature **136** is shown as extending along a circumferential direction of the cylindrical surface cylindrical outer surface **38** of upper steering jacket **18** along a substantial length, or the entirety of the length of scallop **44**, thereby providing maximum resistance to rotation of the mount bracket **20** upon forcibly impacting the surface **43** during an energy absorption event.

(17) In accordance with another aspect of the disclosure, a method of causing a steering jacket **18** to inhibit rotation of a mount bracket, also referred to as mount plate **20**, as the mount plate **20** moves from a releasably fixed position with a fixed vehicle support member **22** to a released position from the fixed vehicle support member **22** during a steering column energy absorbing event, is provided. The method includes, providing the steering jacket **18** having at least one interface feature **36** extending toward the mount plate **20**, with the at least one interface feature **36**

arranged to confront the mount plate **20** during the steering column energy absorbing event to inhibit rotational movement of the mount plate **20** relative to the steering jacket **18** when the mount plate **20** moves from the releasably fixed position to the released position.

(18) The method can further include providing the at least one interface feature **36** as a monolithic piece of material with the steering jacket **18**.

(19) The method can further include providing the at least one interface feature as a plurality of interface features **36**, and further yet, providing the plurality of interface features **36** in circumferentially spaced from one another about an outer surface **38** of the steering jacket **18**. Further, the method can include providing the plurality of interface features **36** in equidistantly spaced relation from an end **40** of the steering jacket **18**.

(20) The method can further include providing the at least one interface feature as a single interface feature **36**, and further, providing the single interface feature **36** as a plastically deformed material of the steering jacket **18**, with the interface feature **36** extending radially outwardly from a generally cylindrical outer surface **38** of the steering jacket **18**.

(21) While the invention has been described in detail in connection with only a limited number of embodiments, it should be readily understood that the invention is not limited to such disclosed embodiments. Rather, the invention can be modified to incorporate any number of variations, alterations, substitutions or equivalent arrangements not heretofore described, but which are commensurate with the spirit and scope of the invention. Additionally, while various embodiments of the invention have been described, it is to be understood that aspects of the invention may include only some of the described embodiments. Accordingly, the invention is not to be seen as limited by the foregoing description.

Claims

1. A steering column assembly, comprising: a mount bracket configured for attachment in a releasably fixed position to a fixed vehicle support member and for release from the fixed vehicle support member to a released position during a steering column energy absorbing event; and a steering jacket operably supported by said mount bracket, said steering jacket having at least one interface feature arranged to confront said mount bracket, during the steering column energy absorbing event, to inhibit rotational movement of said mount bracket relative to said steering jacket when said mount bracket moves from the releasably fixed position to the released position.
2. The steering column assembly of claim 1, wherein said at least one interface feature is a monolithic piece of material of said steering jacket.
3. The steering column assembly of claim 2, wherein said at least one interface feature is plastically deformed material of said steering jacket.
4. The steering column assembly of claim 1, wherein said at least one interface feature extends radially outwardly from a generally cylindrical outer surface of said steering jacket toward said mount bracket.
5. The steering column assembly of claim 1, wherein said at least one interface feature includes a plurality of interface features.
6. The steering column assembly of claim 5, wherein said plurality of interface features are circumferentially spaced from one another.
7. The steering column assembly of claim 6, wherein said plurality of interface features are equidistantly spaced from an end of said steering jacket.
8. The steering column assembly of claim 1, wherein said at least one interface feature is a single interface feature.
9. The steering column assembly of claim 8, wherein said interface feature extends radially outwardly from a generally cylindrical outer surface of said steering jacket toward said mount bracket.

10. The steering column assembly of claim 9, wherein said interface feature extends in a circumferential direction about a portion of said generally cylindrical surface.
 11. The steering column assembly of claim 1, wherein said at least one interface feature is configured to engage an outer peripheral edge of said mount bracket.
 12. The steering column assembly of claim 1, wherein said mount bracket is a rake bracket.
 13. A method of causing a steering jacket to inhibit rotation of a mount plate as the mount plate moves from a releasably fixed position with a fixed vehicle support member to a released position from the fixed vehicle support member during a steering column energy absorbing event, comprising: providing the steering jacket having at least one interface feature extending toward the mount plate, with the at least one interface feature arranged to confront the mount plate during the steering column energy absorbing event to inhibit rotational movement of the mount plate relative to the steering jacket when the mount plate moves from the releasably fixed position to the released position.
 14. The method of claim 13, further including providing the at least one interface feature as a monolithic piece of material with the steering jacket.
 15. The method of claim 13, further including providing the at least one interface feature as a plurality of interface features.
 16. The method of claim 15, further including providing the plurality of interface features in circumferentially spaced relation from one another about an outer surface of the steering jacket.
 17. The method of claim 16, further including providing the plurality of interface features in equidistantly spaced relation from an end of the steering jacket.
 18. The method of claim 15, further including providing the plurality of interface features from plastically deformed material of the steering jacket, with the plurality of interface features extending radially outwardly from a generally cylindrical outer surface of the steering jacket.
 19. The method of claim 13, further including providing the at least one interface feature as a single interface feature.
 20. The method of claim 19, further including providing the interface feature as a plastically deformed material of the steering jacket, with the interface feature extending radially outwardly from a generally cylindrical outer surface of the steering jacket.
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